

Bhushan Ladgaonkar

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PROFESSIONAL SUMMARY

Security Researcher and Machine Learning Engineer specializing in **Adversarial AI** and **Offensive Security**. Experienced in building ML-based Network Intrusion Detection Systems (NIDS) for enterprise infrastructure and developing full-stack intelligence platforms. Seeking to leverage expertise in simulation, modeling, and security architectures.

EDUCATION

Fr. Conceicao Rodrigues College of Engineering
Bachelor of Technology (B.Tech) in Computer Engineering

Mumbai, India
Expected May 2028

EXPERIENCE

Oil and Natural Gas Corporation (ONGC)

Summer Intern | Database & Security Group

Mumbai, India

May 2025 – June 2025

- Built a prototype **Network Intrusion Detection System (NIDS)** on the UNSW-NB15 dataset (2M+ records) to detect zero-day threats; used **SMOTE** for class-imbalance handling and advanced feature engineering.
- Implemented **Random Forest** and **XGBoost** classifiers, reducing false positive rates while sustaining high detection sensitivity for anomalous network packets.
- Documented threat detection methodologies in collaboration with the Database Group, translating raw data findings into actionable security intelligence.

TECHNICAL PROJECTS

Energy Crisis & Sustainability Tracker | *React, Node.js, MongoDB, LLaMA 3.3, WebSockets*

May 2026

- Collaborated to engineer a real-time intelligence platform tracking global energy markets and SDG targets, integrating a domain-restricted AI chatbot powered by **Groq SDK (LLaMA 3.3-70B)**.
- Developed a scalable backend using **Express.js** and **MongoDB**, implementing **Socket.io** for live price broadcasting and **node-cron** for automated data ingestion from Yahoo Finance and IEA.
- Built a responsive, HUD-themed frontend with **React 19**, **TypeScript**, and **Tailwind CSS**, featuring an interactive D3/SVG world heatmap to visualize energy indices and geopolitical conflict impacts.

TISD-Edu-AI: Local-First RAG Tutor | *Python, MLX, ChromaDB, FastAPI, LLMs*

Apr 2026

- Engineered a privacy-focused, on-device **RAG** pipeline using **Microsoft Phi-3 (3.8B)** and **ChromaDB**, achieving **41.42 tokens/sec** at 1.72s latency with a 9.99 GB memory footprint on Apple Silicon.
- Ingested 4,600+ pages of academic PDFs into 6,400+ semantic chunks; validated with a custom eval suite achieving **81.1% semantic accuracy** across a 100-sample stress test.

PUBLICATIONS

Benchmarking Resilience: Asymmetric Latent Purification...

Accepted: May 2026

First & Corresponding Author. Intl. Journal of Research and Innovation in Applied Science.

Mumbai, India

- Engineered the **Asymmetric Latent Purifier (ALP)**, a zero-shot structural defense against iterative PGD attacks utilizing an ultra-compact 64-channel stochastic bottleneck and a **ResNet-Lite** backbone.
- Implemented a 20-sample **Monte Carlo Ensemble** on an Apple M4 GPU, recovering **CIFAR-10** classification accuracy to **32.06%** (25.68 dB PSNR) under severe threat.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, C, SQL, Java

AI & ML: Machine Learning, RAG, LLMs, PyTorch, MLX, ChromaDB, XGBoost

Web & Backend: React 19, Node.js, Express.js, MongoDB, Socket.io, Tailwind CSS, Docker, FastAPI

Security & Tools: Network Intrusion Detection (NIDS), Offensive Security, OSINT, Linux, Git, VS Code

LEADERSHIP & CERTIFICATIONS

Institution's Innovation Council (IIC) – FRCRCE

Mumbai, India

IPR Activities Coordinator

Apr 2026 – Present

- Coordinating IPR, patent filing, and research commercialization initiatives to foster an innovation ecosystem among engineering students.

McKinsey & Company

Remote

Selected Participant – McKinsey Forward Program

May 2026 – Present

- Completed a professional development program focused on structured problem-solving, adaptability, and digital communication strategies for business-technology integration.

CS50 Cybersecurity – Harvard University

Offensive Security Operations – Cybrary

OSINT – Cybrary

Cyber Kill Chain – Cybrary